

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Dot .Net Core and Progressive Web Application Practical

Practical – IV

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Part A - Theory:

1. What is ASP.NET Web Forms?

ASP.NET Web Forms is a web application framework developed by Microsoft for building dynamic, interactive, and data-driven websites. It provides a rich collection of **server controls** that simplify web application development by reducing the amount of code required.

Features of ASP.NET Web Forms:

- Event-driven programming model.
- Rich collection of server controls.
- Built-in state management using ViewState.
- Easy integration with databases.
- Support for validation controls.
- Rapid application development.

2. What are Server Controls?

Server Controls are controls that run on the server and generate HTML dynamically. They simplify user interaction and automatically handle client-server communication.

Syntax:

```
<asp:ControlName ID="ControlID" runat="server" />
```

3. Common ASP.NET Server Controls

a) Label Control

The **Label** control is used to display static text or messages on a web page.

Syntax

```
<asp:Label ID="lblName" runat="server" Text="Student Name"></asp:Label>
```

b) TextBox Control

The **TextBox** control allows the user to enter text.

Syntax

```
<asp:TextBox ID="txtName" runat="server"></asp:TextBox>
```

c) RadioButton Control

The **RadioButton** control allows the user to select only one option from a group.

Syntax

```
<asp:RadioButton ID="rbMale" runat="server" Text="Male" GroupName="Gender" />
```

d) DropDownList Control

The **DropDownList** control displays a list of options from which the user can select one.

Syntax

```
<asp:DropDownList ID="ddlCourse" runat="server">  
    <asp:ListItem>BSc IT</asp:ListItem>  
    <asp:ListItem>BSc CS</asp:ListItem>  
    <asp:ListItem>BCA</asp:ListItem>  
</asp:DropDownList>
```

e) CheckBox Control

The **CheckBox** control allows the user to select or deselect an option.

Syntax

```
<asp:CheckBox ID="chkTerms" runat="server" Text="I Agree" />
```

f) Calendar Control

The **Calendar** control allows users to select a date from a graphical calendar.

Syntax

```
<asp:Calendar ID="Calendar1" runat="server"></asp:Calendar>
```

g) Button Control

The **Button** control is used to perform an action when clicked.

Syntax

```
<asp:Button ID="btnSubmit" runat="server" Text="Submit" OnClick="btnSubmit_Click" />
```

4. Validation Controls

Validation controls are used to ensure that the user enters valid data before submitting a form.

Types of Validation Controls

a) RequiredFieldValidator

Ensures that a control is not left empty.

```
<asp:RequiredFieldValidator  
ID="rfvName"  
runat="server"  
ControlToValidate="txtName"
```

```
ErrorMessage="Name is required">  
</asp:RequiredFieldValidator>
```

b) RangeValidator

Checks whether a value lies within a specified range.

c) CompareValidator

Compares the value of one control with another control or a fixed value.

d) RegularExpressionValidator

Validates the input using a regular expression (e.g., Email, Mobile Number).

e) CustomValidator

Allows custom validation logic written by the programmer.

f) ValidationSummary

Displays all validation error messages in one place.

5. AdRotator Control

The **AdRotator Control** displays different advertisements or images randomly from an XML file. It is commonly used to display advertisements, announcements, offers, or promotional banners.

Advantages

- Displays advertisements automatically.
- Randomly rotates images.
- Reads advertisement details from an XML file.
- Easy to maintain and update.

Syntax

```
<asp:AdRotator  
ID="AdRotator1"  
runat="server"  
AdvertisementFile="Ads.xml">  
</asp:AdRotator>
```

6. Advantages of ASP.NET Server Controls

- Easy to use and configure.
- Supports event-driven programming.
- Reduces coding effort.
- Provides automatic state management.
- Supports built-in validation.
- Improves application security.
- Generates browser-compatible HTML automatically.

7. Applications

- Student Registration System
- Online Admission Portal
- Employee Registration System
- Online Examination System
- Library Management System
- E-Commerce Registration Forms
- Hospital Appointment System

Part B - Program

a) Create an ASP.NET Web Forms application using simple server controls.

a. Design a Student Registration Form using the following ASP.NET controls:

- i. Label and TextBox to accept Student Name
- ii. RadioButton to select Gender (Male/Female)
- iii. DropDownList to select Course (BSc IT, BSc CS, BCA)
- iv. CheckBox for accepting Terms & Conditions
- v. Calendar control to select Date
- vi. Button control to submit the form
- vii. Label control to display the output

Algorithm:

Step 1: Start the program.

Step 2: Create the registration form using [ASP.NET](#) server controls.

Step 3: Accept user inputs (Name, Gender, Course, Date, etc.).

Step 4: Click the **Submit** button to display the entered details.

Step 5: Stop the program.

Code:

```
<%@ Page Title="Home Page" Language="C#" MasterPageFile="~/Site.Master"
AutoEventWireup="true" CodeFile="Default.aspx.cs" Inherits="_Default" %>
```

```
<asp:Content ID="BodyContent" ContentPlaceHolderID="MainContent" runat="server">
```

```
<div>
```

```
<br />
```

```
Student name :-<asp:TextBox ID="TextBox1" runat="server"></asp:TextBox>
```

```
<br />
```

```
<asp:RadioButtonList ID="RadioButtonList1" runat="server">
```

```
<asp:ListItem Value="M">Male</asp:ListItem>
```

```
<asp:ListItem Value="F">Female</asp:ListItem>
```

```
<asp:ListItem Value="O">Others</asp:ListItem>
```

```

</asp:RadioButtonList>
<br />
<asp:DropDownList ID="DropDownList1" runat="server" Height="36px"
OnSelectedIndexChanged="DropDownList1_SelectedIndexChanged" Width="125px">
    <asp:ListItem Value="s">Select a option</asp:ListItem>
    <asp:ListItem>BSC IT</asp:ListItem>
    <asp:ListItem>BSC CS</asp:ListItem>
    <asp:ListItem>BCA</asp:ListItem>
    <asp:ListItem></asp:ListItem>
</asp:DropDownList>
<br />
<asp:CheckBox ID="CheckBox1" runat="server" Text="I accept Terms and Conditions" />
<br />
<asp:Calendar ID="Calendar1" runat="server"></asp:Calendar>
<br />
<asp:Button ID="Button1" runat="server" OnClick="Button1_Click" Text="Submit" />
<br />
<asp:Label ID="Label1" runat="server" Text="Label"></asp:Label>

</div>
</asp:Content>

```

Submit:

```

string name = TextBox1.Text;
string gender = RadioButtonList1.Text;
string course = DropDownList1.Text;
string date = Calendar1.SelectedDate.ToString();
Label1.Text = name + "<br/>" + gender + "<br/>" + course + "<br/>" + date;

```

Output:

Student name :- bhavesh

☒ Male

☐ Female

☐ Others

BSC IT

☒ I accept Terms and Conditions

July 2026						
Sun	Mon	Tue	Wed	Thu	Fri	Sat
28	29	30	1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	1
2	3	4	5	6	7	8

Submit

bhavesh

M

BSC IT

09-07-2026 00:00:00

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b) Create a Registration form to demonstrate use of various Validation controls.

Algorithm:

Step 1: Start the program.

Step 2: Create the registration form with input fields.

Step 3: Apply validation controls (RequiredFieldValidator, RangeValidator, CompareValidator).

Step 4: Validate the entered data when the button is clicked.

Step 5: Stop the program.

Code:

```
<%@ Page Title="Home Page" Language="C#" MasterPageFile="~/Site.Master" AutoEventWireup="true"
CodeFile="Default.aspx.cs" Inherits="_Default" %>
```

```
<asp:Content ID="BodyContent" ContentPlaceHolderID="MainContent" runat="server">
```

```
<div>
```

```
<br />
```

```
<asp:Label ID="Label1" runat="server" Text="Name"></asp:Label>
```

```
<asp:TextBox ID="TextBox1" runat="server"></asp:TextBox>
```

```
<asp:RequiredFieldValidator ID="RequiredFieldValidator1" runat="server" ControlToValidate="TextBox1"
ErrorMessage="name should not be empty"></asp:RequiredFieldValidator>
```

```
<br />
```

```
<asp:Label ID="Label2" runat="server" Text="age"></asp:Label>
```

```
<asp:TextBox ID="TextBox2" runat="server"></asp:TextBox>
```

```
<asp:RangeValidator ID="RangeValidator1" runat="server" ControlToValidate="TextBox2" ErrorMessage="Age
should be 18 or 99" MaximumValue="99" MinimumValue="18"></asp:RangeValidator>
```

```
<br />
```

```

<asp:Label ID="Label3" runat="server" Text="Password"></asp:Label>
<asp:TextBox ID="TextBox3" runat="server"></asp:TextBox>
<br />
<asp:Label ID="Label4" runat="server" Text="Confirm Password"></asp:Label>
<asp:TextBox ID="TextBox4" runat="server"></asp:TextBox>
<asp:CompareValidator ID="CompareValidator1" runat="server" ControlToCompare="TextBox3"
ControlToValidate="TextBox4" ErrorMessage="password not match"></asp:CompareValidator>
<br />
<asp:Button ID="Button1" runat="server" Text="Button" />
</div>
</asp:Content>

```

Output:

CSSClass	
Display	Static
ErrorMessage	name should not be empty
Font	
ForeColor	
Text	
Behavior	
ClientIDMode	Inherit
ControlToValidate	TextBox1
EnableClientScript	True
Enabled	True

Display	Static
ErrorMessage	Age should be 18 or 99
Font	
ForeColor	
Text	
Behavior	
ClientIDMode	Inherit
ControlToValidate	TextBox2
CultureInvariantVa	False
EnableClientScript	True
Enabled	True
EnableTheming	True
EnableViewState	True
MaximumValue	99
MinimumValue	18

Display	Static
ErrorMessage	password not match
Font	
ForeColor	
Text	
Behavior	
ClientIDMode	Inherit
ControlToCompare	TextBox3
ControlToValidate	TextBox4
CultureInvariantV	False
EnableClientScript	True
Enabled	True

c) Create Web Form to demonstrate use of Adrotator Control.

Algorithm:

Step 1: Start the program.

Step 2: Create an advertisement XML file with image and URL details.

Step 3: Add the AdRotator control to the web form.

Step 4: Display the advertisement and redirect to the specified URL on click.

Step 5: Stop the program.

Code:

```
<asp:AdRotator ID="AdRotator1" runat="server" AdvertisementFile="~/XMLFile.xml" />
<?xml version="1.0" encoding="utf-8" ?>
<Advertisements>
  <Ad>
    <ImageUrl>Images/sun.jpg</ImageUrl>
    <AlternateText>sunflower</AlternateText>
    <Height>50</Height>
  </Ad>
</Advertisements>
```

Output:

Name

age

Password

Confirm Password



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Part C – Curiosity Question

1. Online College Admission System

Develop an **ASP.NET Web Forms application** for an **Online College Admission System**. The application should allow students to register by entering their personal and academic details using appropriate **server controls**. Validate all mandatory fields using suitable **Validation Controls** and display meaningful error messages. After successful registration, display a confirmation message with the entered details. Additionally, use an **AdRotator Control** to display rotating banners for college events, admissions, scholarships, or placement drives.